

AEROSPACE

PROBLEM STATEMENT ID	PROBLEM STATEMENT	PROBLEM STATEMENT DESCRIPTION
AERO-01	Orbital Debris Conjunction Alert System Using Open TLE Data	Develop a Space Debris Tracking and Conjunction Alert System that uses live TLE data to propagate satellite orbits (SGP4), detect potential collisions within a user-defined threshold over 72 hours, estimate collision probability, and recommend minimum delta-v avoidance maneuvers. The system includes 2D/3D orbit visualization and provides real-time conjunction alerts for affordable space traffic management.
AERO-02	Terrain-Aware Autonomous UAV Path Planner for Agricultural Surveys	Develop an AI-based UAV Path Planning System that uses GeoTIFF terrain maps and field boundaries to generate obstacle-aware, energy-optimized drone survey paths using A^* / RRT^* algorithms. The system adapts to wind conditions, estimates energy consumption, and exports MAVLink-compatible waypoint files with 2D path visualization, enabling efficient agricultural drone operations in hilly terrain.
AERO-03	Satellite Change Detection System for Airport OLS Zone	Develop an AI-based Satellite Change Detection System that compares multi-

	Encroachment Monitoring	temporal Sentinel-2/Bhuvan satellite images to detect new built-up areas using U-Net or image differencing. The system identifies change polygons with coordinates and severity, checks them against AAI Obstacle Limitation Surface (OLS) boundaries, and generates a downloadable alert report with map visualization for monitoring unauthorized construction near airports.
AERO-04	Drone Swarm Coordination Simulator for Disaster Search & Rescue	Develop an AI-based Multi-Drone Swarm Simulation for disaster search and rescue that simulates multiple drones using a decentralized coordination algorithm (e.g., stigmergy, auction-based, or MARL) to maximize area coverage and victim detection under battery constraints. The system provides swarm path visualization, evaluates coverage and detection metrics, and compares performance against a single-drone baseline.
AERO-05	AI Space Weather Impact Predictor for LEO Satellite Operations	Develop an AI-based Space Weather Impact Prediction System for LEO satellites that uses NOAA SWPC alerts and Kp index data to predict atmospheric drag and orbit decay using LSTM/Gaussian Process models. The system estimates re-entry timeline changes for user-defined satellite TLEs, displays a risk dashboard, and sends configurable alerts when

		space weather conditions exceed specified Kp thresholds.
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